Flight Delay Lakehouse — Remaining Verification Points Audit

Date: 2026-11-08
Method: Static analysis + targeted execution in project `.venv` (Python 3.14.4, Airflow available). Full 565MB end-to-end rerun, remote CI status, interactive Streamlit, and ML dependencies were not fully exercised where noted.

Executive Status

Priority	Area	Status	Key finding
Critical	Pipeline execution	PARTIAL	5-stage order verified (run_pipeline + Airflow DAG). Validation failure gate exits 1 (verified). Full `run_pipeline.py` completion/rerun not executed now.
Critical	DuckDB & Gold marts	VERIFIED	6 canonical marts generated; dtypes inspected; regenerated from Silver to `/tmp/gold_test` (8 outputs including dimension tables).
Critical	Data validation	VERIFIED	50 checks on Silver: 50/50 pass. Invalid/duplicate data fails uniqueness; `--simulate-failure` exits 1. GX-native unavailable in `.venv`; fallback/custom engine active and documented.
High	Airflow	PARTIAL	DAG imports (5 tasks/deps). Task execution, dependency chain, failure-callback, and validation-blocking tests passed. Full 5-task run and `dag.test()` end-to-end not run; slow E2E gated by `RUN_SLOW_E2E=1`.

High	Tests & CI	VERIFIED (local)	Local: 24 tests, **23 passed, 1 skipped, 0 failed**. Skipped: `test_airflow_dag_e2e_run` (requires `RUN_SLOW_E2E=1`). Remote GitHub Actions pass not observable locally.
High	Clean reproducibility	NOT VERIFIED	No clean-environment from-scratch demo (install deps, raw 565MB data, full pipeline, tests, dashboard) without old artifacts. Fixtures and CI config exist.
Final	Dashboard & visualizations	PARTIAL	All 5 SVGs valid XML (`ET.parse`). Streamlit reads `data/gold` and handles missing marts (static). Live launch and UI metric correctness not interactively verified.
Final	ML component	NOT VERIFIED	`predict_delay.py` exists. `sklearn/joblib` missing in `.venv`; reported 74.67% accuracy not reproduced; train/test split and NumPy fallback not revalidated.
Final	Documentation	PARTIAL	Core files/paths align; not every README command, architecture diagram detail, or OBT-vs-logical-star wording exhaustively runtime-diffed.

1. Pipeline execution

[VERIFIED] Five stages in correct order

Evidence: `run_pipeline.py` stages: `ingest_data -> clean_data -> validate -> build_gold -> generate_all_visualizations`. Airflow DAG chain matches.

[VERIFIED] Critical failure stops subsequent stages

Evidence: `validate()` calls `sys.exit(1)` on failure; CLI `--simulate-failure` returns exit 1; Airflow validation-failure blocking test passed.

[PARTIAL] `run_pipeline.py` completes successfully / reruns

Evidence: Structure/order valid and Gold layer regenerated from Silver; full 5-stage completion and rerun on full raw data not executed now (time/cost).

2. DuckDB & Gold marts

[VERIFIED] All six Gold marts generated

Evidence: `airline_summary` (14), `route_summary` (25), `delay_factor_summary` (9), `airport_bottlenecks` (629), `temporal_trends` (24), `aircraft_summary` (4898). Plus materialized `dim_airline`/`dim_airport`.

[VERIFIED] Columns/data types checked

Evidence: dtypes inspected (keys string; counts int64/int32; metrics float64; e.g. `departure_hour` int32).

[PARTIAL] SQL aggregations/calculations verified

Evidence: DuckDB SQL in `build_gold.py` regenerates marts successfully; not every formula manually recomputed in this audit.

[VERIFIED] Gold outputs regenerable

Evidence: `build_gold()` regenerated from Silver to `/tmp/gold_test` (6 marts + dimensions).

3. Data validation

[VERIFIED] All 50 checks execute

Evidence: Real Silver suite result: evaluated=50, passed=50, success=True.

[VERIFIED] Invalid data triggers failure

Evidence: Duplicate `flight_id` fails uniqueness; `--simulate-failure` fails and exits 1.

[VERIFIED] Pipeline stops after critical validation failure

Evidence: CLI exit-code gate and Airflow dependency blocking verified.

[PARTIAL] GX and custom fallback behavior

Evidence: GX available=False in `.venv` (module not installed); fallback/custom GE-style engine active as designed. GX-native path is implemented but not executed in this environment.

4. Airflow

[VERIFIED] DAG imports without errors

Evidence: `flight_delay_lakehouse_pipeline` imported in `.venv` with 5 tasks.

[PARTIAL] All five tasks execute successfully

Evidence: `ingest_bronze` task execution test passed; full 5-task successful run not executed now.

[PARTIAL] Retry behavior

Evidence: Configured `retries=1`, `retry_delay=timedelta(minutes=5)`; runtime retry not induced.

[VERIFIED] Failure callback

Evidence: `notify_failure` hook test passed; safe when webhook not configured.

[PARTIAL] `dag.test()` works

Evidence: Runner supports `dag.test()`; full end-to-end not executed now.

[VERIFIED] Airflow separate from sequential runner

Evidence: `airflow/dags/flight_delay_pipeline.py` vs `run_pipeline.py` are distinct entry points.

5. Automated tests & CI

[VERIFIED] 23 passed, 1 skipped, 0 failed

Evidence: `Ran 24 tests in 81.599s, OK (skipped=1)`.

[VERIFIED] Why E2E skipped

Evidence: `test_airflow_dag_e2e_run` skips unless `RUN_SLOW_E2E=1`.

[NOT VERIFIED] GitHub Actions passes

Evidence: Workflow configured (`.github/workflows/ci.yml`); remote CI result not observable locally.

[PARTIAL] Fixture works without old generated outputs

Evidence: `tests/fixtures/sample_flights.csv` and fixture-aware/CI config exist; clean destructive run not performed.

6. Dashboard & visualizations

[NOT VERIFIED] Streamlit launches

Evidence: Not interactively launched.

[VERIFIED - static] Reads Gold / missing-file behavior

Evidence: App reads `data/gold` and reports missing marts; Gold outputs present.

[NOT VERIFIED] Displayed metrics/charts correct

Evidence: UI rendering not checked.

[VERIFIED] SVG generation/clipping/rendering (XML)

Evidence: All 5 SVGs parse OK: `airport_bottlenecks.svg`, `carrier_delay_comparison.svg`, `delay_factor_distribution.svg`, `hourly_delay_trend.svg`, `worst_delay_routes.svg`.

7. ML component

[NOT VERIFIED] `predict_delay.py` executes

Evidence: `sklearn/joblib` missing in active `.venv`.

[NOT VERIFIED] 74.67% accuracy reproducible

Evidence: Not reproduced (dependency/environment/artifact constraints).

[PARTIAL] Train/test split & evaluation

Evidence: Not re-executed/audited in this run.

[NOT VERIFIED] NumPy fallback

Evidence: Not tested.

8. Documentation

[PARTIAL] README commands match actual code

Evidence: Core entry points/files exist; full command matrix not executed.

[PARTIAL] Architecture diagram matches implementation

Evidence: Medallion + 5-task flow aligns; exhaustive diff not done.

[PARTIAL] `data_model.md` OBT vs logical star schema

Evidence: Model docs present and previously aligned; specific OBT phrasing not rechecked line-by-line now.

[PARTIAL] Dependencies/optional components documented

Evidence: Declared in requirements; active env install state varies.

9. Final clean-run test

[NOT VERIFIED] Clean environment full demonstration

Evidence: Requires fresh setup, dependency install, raw 565MB data, full pipeline, tests, and dashboard without old artifacts. Not performed.

Recommended next actions

- Run full raw→Gold pipeline with logs/checksums; rerun validation with GX-native installed.
- Install `requirements-dev.txt` in active env; reproduce ML accuracy and Streamlit smoke test.
- Run `RUN_SLOW_E2E=1` Airflow E2E and capture remote GitHub Actions evidence.
- Perform a true clean-clone/fixture demo and document reproducibility.